

Section 1. Chemical product and company identification

Chinese name : 濃縮洗衣粉
Product name English : LAUNDRI PERMA BRITE
Supplier : Ecolab Limited
 15/F, Lu Plaza
 2 Wing Yip Street, Kwun Tong, Kowloon, HONG KONG
 Tel: +852-2341 4202
 Main office: 1588-7595
 Plant: 1544-7686
 Responsible name: Korea Technical Center
Emergency telephone number : 852-2341-4202
Product type : Laundry detergent
Date of issue : 2011 . 5 . 26

Section 2. Hazards identification

Hazard classification : Class 8 Corrosive substances
Routes of entry : Skin contact, Eye contact, Ingestion
Human health hazards
Inhalation : Severely irritating to the respiratory system.
Ingestion : Causes burns to mouth, throat and stomach. May be harmful if swallowed.
Skin : Corrosive to the skin.
Eyes : Corrosive to eyes.
Environmental hazards : No known significant effects or critical hazards.
Fire/explosion hazards : Fine dust clouds may form explosive mixtures with air.

Section 3. Composition/information on ingredients

Mixture

Chemical name	Concentration	CAS No.
sodium carbonate	>50	497-19-8
disodium metasilicate	10 - 30	6834-92-0
triphosphoric acid, pentasodium salt	1 - 10	7758-29-4
alcohols, c12-16, ethoxylated	1 - 10	68551-12-2

Section 4. First-aid measures

Inhalation : If inhaled, remove to fresh air. If exposed person is not breathing, give artificial respiration or oxygen applied by trained personnel. Get medical attention immediately.
Ingestion : If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Do not induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.
Skin contact : In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Clean shoes thoroughly before reuse. Get medical attention immediately. Wash clothing before reuse.
Eye contact : In case of contact, immediately flush eyes with cool running water. Remove contact lenses and continue flushing with plenty of water for at least 15 minutes. Get medical attention immediately.

Section 5. Fire-fighting measures

Extinguishing media

Suitable	: Use dry chemical powder.
Not suitable	: Do not use water jet.
Hazardous combustion products	: Decomposition products may include the following materials: carbon dioxide carbon monoxide phosphorus oxides metal oxide/oxides
Special protective equipment for fire-fighters	: Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.
Materials to avoid	acids

Section 6. Accidental release measures

Personal precautions	: Ventilate area of leak or spill. Do not touch damaged containers or spilled material unless wearing appropriate protective equipment (Section 8). Stop leak if without risk. Do not allow to enter drains or watercourses.
Environmental precautions	: Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).
Methods for cleaning up	: If emergency personnel are not present, contain and collect small spillages with non-combustible absorbent material, e.g. sand, earth or other suitable material and place in a suitable container for disposal. Use a water rinse for final clean-up.

Section 7. Handling and storage

Handling	: Do not ingest. Do not get in eyes or on skin or clothing. Use only with adequate ventilation. Keep container closed. Wash thoroughly after handling.
Storage	: Keep out of reach of children. Keep container tightly closed. Keep container in a cool, well-ventilated area. Do not store above the following temperature: 45°C

Section 8. Exposure controls/personal protection

Exposure limit values	: Not available.
Recommended monitoring procedures	: Not applicable.
Occupational exposure controls	: Use only with adequate ventilation. If user operations generate dust, fumes, gas, vapour or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. The engineering controls also need to keep gas, vapour or dust concentrations below any lower explosive limits. Use explosion-proof ventilation equipment.
Hygiene measures	: Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Personal protection

Respiratory	Wear appropriate respirator when ventilation is inadequate and occupational exposure limits are exceeded.
Hands	: Use chemical-resistant, impervious gloves.
Eyes	: Use chemical splash goggles. For continued or severe exposure wear a face shield over the goggles.
Skin	: Use synthetic apron, other protective equipment as necessary to prevent skin contact.

Section 9. Physical and chemical properties

Physical state	: Solid [Powder]
Colour	: White [Light]
Odour	: Odourless
pH	: 12 to 12.5 [Conc. (% w/w): 1%]
Specific gravity	: 0.9 to 1
Solubility	: Partially soluble in the following materials: cold water and hot water.

Section 10. Stability and reactivity

Stability and reactivity	: The product is stable.
Conditions to avoid	: Avoid the creation of dust when handling and avoid all possible sources of ignition (spark or flame). Take precautionary measures against electrostatic discharges. To avoid fire or explosion, dissipate static electricity during transfer by earthing and bonding containers and equipment before transferring material. Prevent dust accumulation.
Materials to avoid	: acids
Hazardous decomposition products	: Under normal conditions of storage and use, hazardous decomposition products should not be produced.

Section 11. Toxicological information**Acute toxicity**

Inhalation	: Severely irritating to the respiratory system.
Ingestion	: Causes burns to mouth, throat and stomach. May be harmful if swallowed.
Skin	: Corrosive to the skin.
Eyes	: Corrosive to eyes.

Potential chronic health effects

Chronic effects	: No known significant effects or critical hazards.
Carcinogenicity	: No known significant effects or critical hazards.
Mutagenicity	: No known significant effects or critical hazards.
Teratogenicity	: No known significant effects or critical hazards.
Developmental effects	: No known significant effects or critical hazards.
Fertility effects	: No known significant effects or critical hazards.

Section 12. Ecological information**Aquatic ecotoxicity**

Product/ingredient name	Result	Species	Exposure
disodium metasilicate	Acute EC50 4857 mg/L	Daphnia	48 hours
	Acute LC50 3185 mg/L	Fish	96 hours
	Acute LC50 2320 mg/L	Fish	96 hours
sodium carbonate	Acute EC50 347 mg/L	Daphnia	48 hours
	Acute EC50 199.82 to 298.9 mg/L Fresh water	Daphnia - Water flea - Neonate - 24 hours	48 hours
	Acute LC50 740 mg/L	Fish	96 hours
	Acute LC50 <850 mg/l	Fish	96 hours
	Acute LC50 320 mg/l	Fish	96 hours
	Acute LC50 300 mg/l	Fish	96 hours
	Acute LC50 1640000 to 2030000 ug/L Fresh water	Daphnia - Water flea - 24 hours	48 hours
	Acute LC50 1020000 to 1170000 ug/L Fresh water	Daphnia - Water flea - 24 hours	48 hours
	Acute LC50 <850000 ug/L Fresh water	Fish - Fathead minnow - 1 to 7 days	96 hours
	Acute LC50 740000 ug/L Fresh water	Fish - Western mosquitofish - Adult	96 hours

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	Acute LC50 565000 ug/L Fresh water	Daphnia - Water flea	48 hours
	Acute LC50 320000 ug/L Fresh water	Fish - Bluegill	96 hours
	Acute LC50 300000 ug/L Fresh water	Fish - Bluegill - 3.88 cm - 0.96 g	96 hours
	Acute LC50 265000 ug/L Fresh water	Daphnia - Water flea	48 hours
triphosphoric acid, pentasodium salt	Acute EC50 >100 mg/L	Daphnia	48 hours
	Acute LC50 >1600 mg/L	Fish	96 hours
	Acute LC50 >100 mg/L	Fish	96 hours

Other adverse effects : No known significant effects or critical hazards.

Environmental effects : No known significant effects or critical hazards.

Section 13. Disposal considerations

Waste disposal : The generation of waste should be avoided or minimised wherever possible. Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe way. Significant quantities of waste product residues should not be disposed of via the foul sewer but processed in a suitable effluent treatment plant. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers.

Consult your local or regional authorities.

Section 14. Transport information

UN number : UN3253
Proper shipping name : DISODIUM TRIOXOSILICATE
Package mark : Corrosive substances
Packing group : III

Section 15. Regulatory information

Authored with regard to:

- "Classification and Mark for commonly used dangerous chemicals" (GB13690-1992)
- "General rules for preparation of a Chemical Safety Data Sheet" (GB16483-2000)
- "The principle of classification of transport packaging groups of dangerous goods" (GB/T15098-94)
- "Classification and code of dangerous goods" (GB6944-2005)

Section 16. Other information

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☑ Indicates information that has changed from previously issued version.

Notice to reader

The above information is believed to be correct with respect to the formula used to manufacture the product in the country of origin. As data, standards, and regulations change, and conditions of use and handling are beyond our control, NO WARRANTY, EXPRESS OR IMPLIED, IS MADE AS TO THE COMPLETENESS OR CONTINUING ACCURACY OF THIS INFORMATION.